V1.0

初心高于胜负

成长胜于输赢



RoboMaster2027

南海鲨电控组

用CUBEMX创建

第一个嵌入式工程

海南大学南海鲨战队 编

2026 年 9 月 7 日

学习目录

1. 选择芯片并创建工程

2. 完成基础配置

3. 配置并生成工程代码

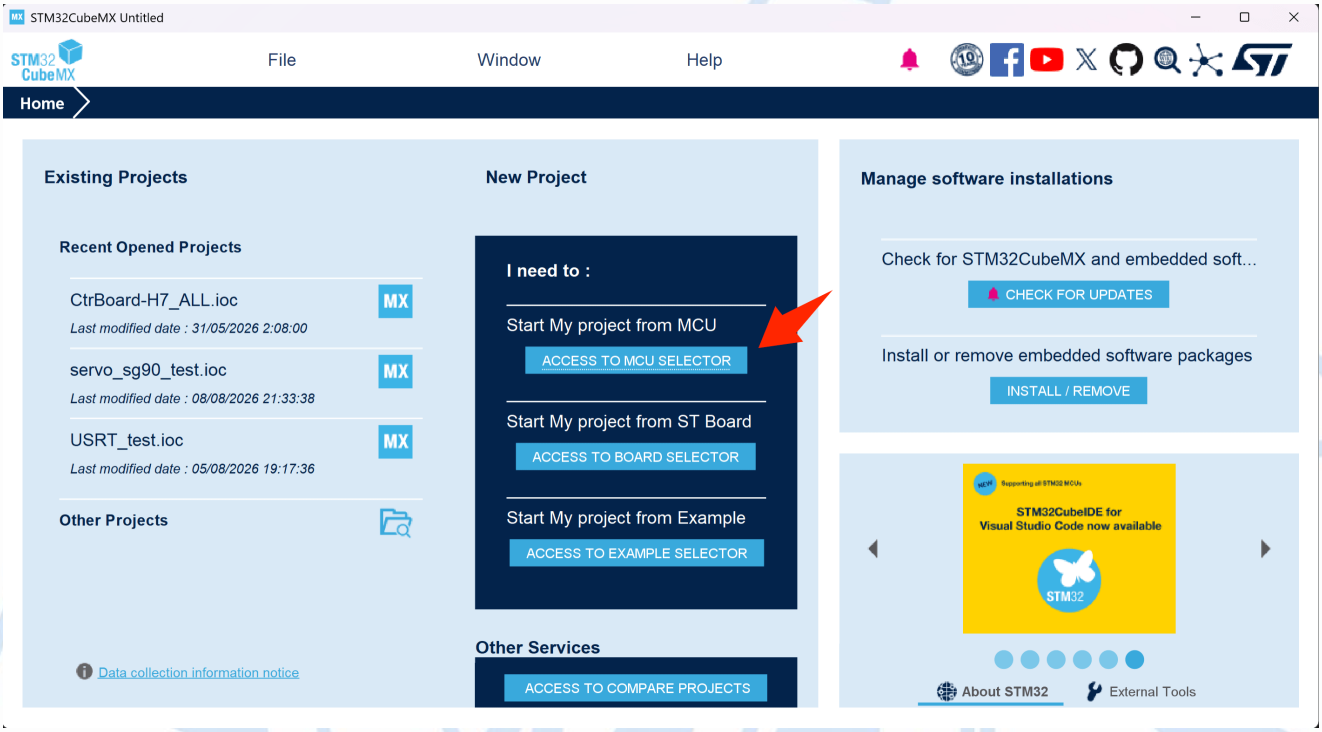
4. 在 CLion 中打开并配置工程

5. 配置 OpenOCD 并编译下载

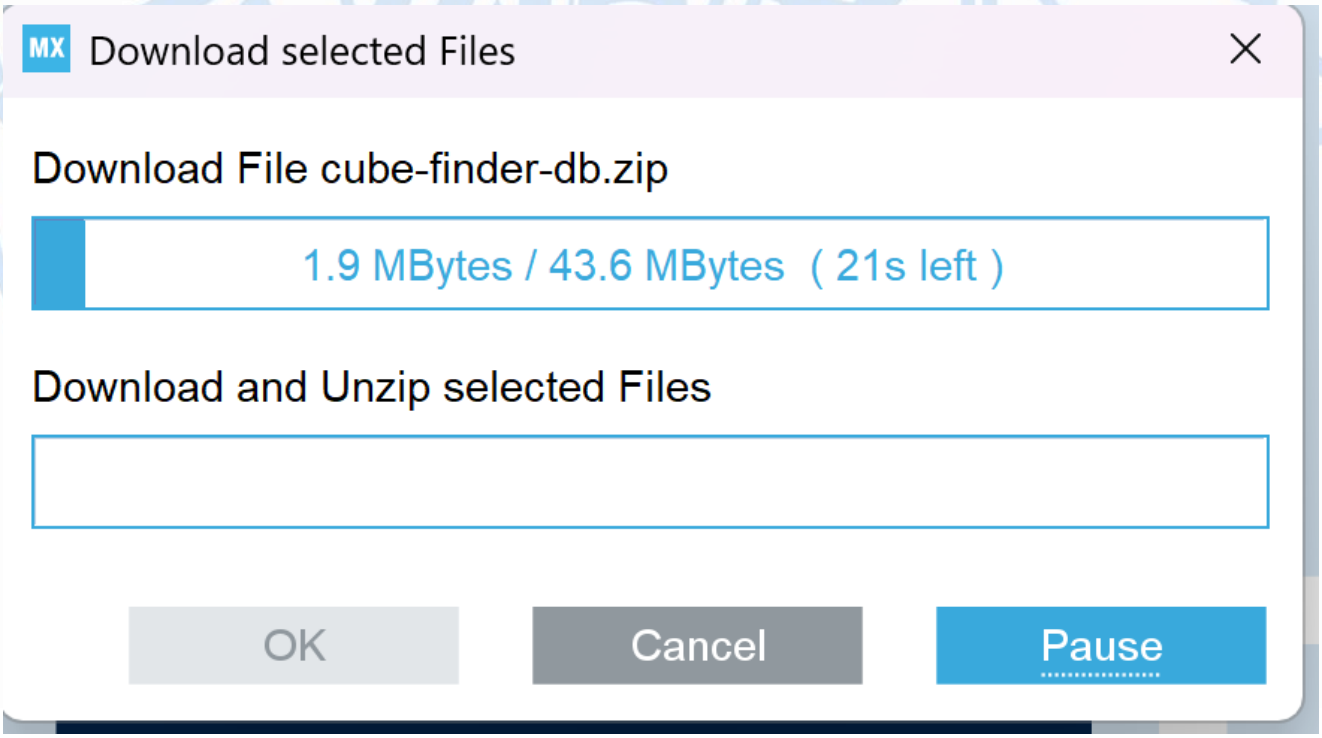
成为伟大点灯工程师的第一步

1. 选择芯片并创建工程

1.1 打开 MCU 选择器



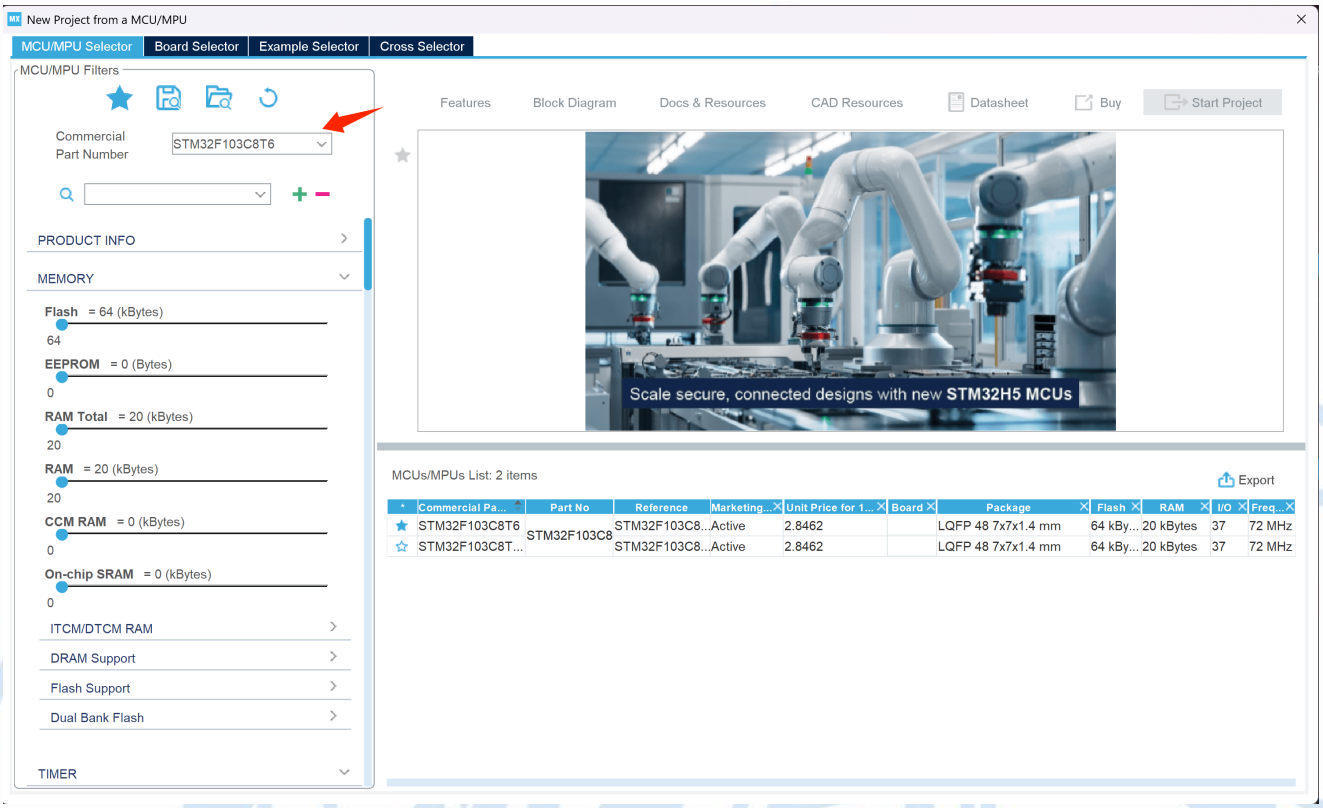
选“ACCESS TO MCU SELECTOR”，这里首次打开会需要下载一堆东西

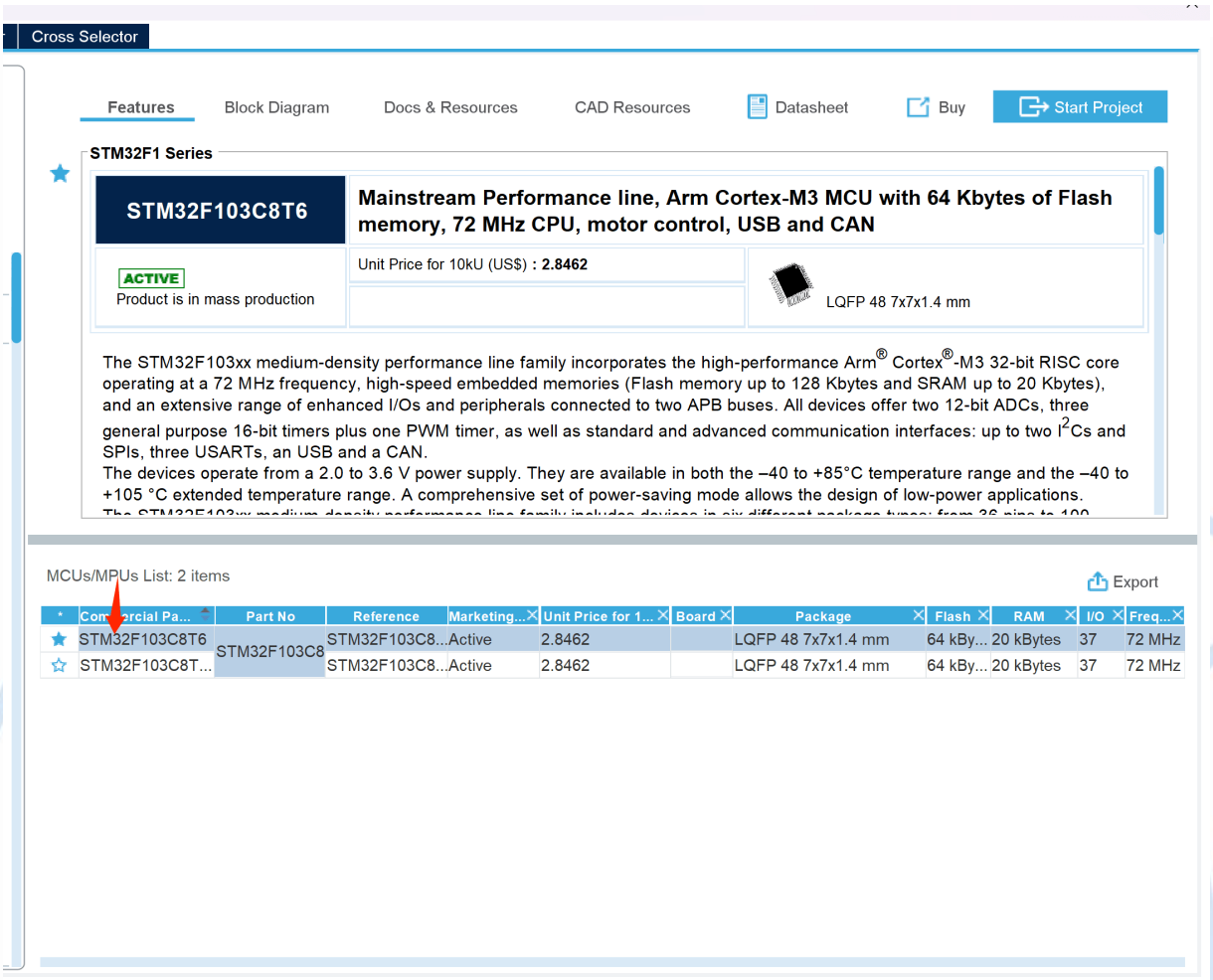


首次使用需要挂梯子下载一次，之后可以直接点cancel不用多次更新

1.2 搜索并选择芯片

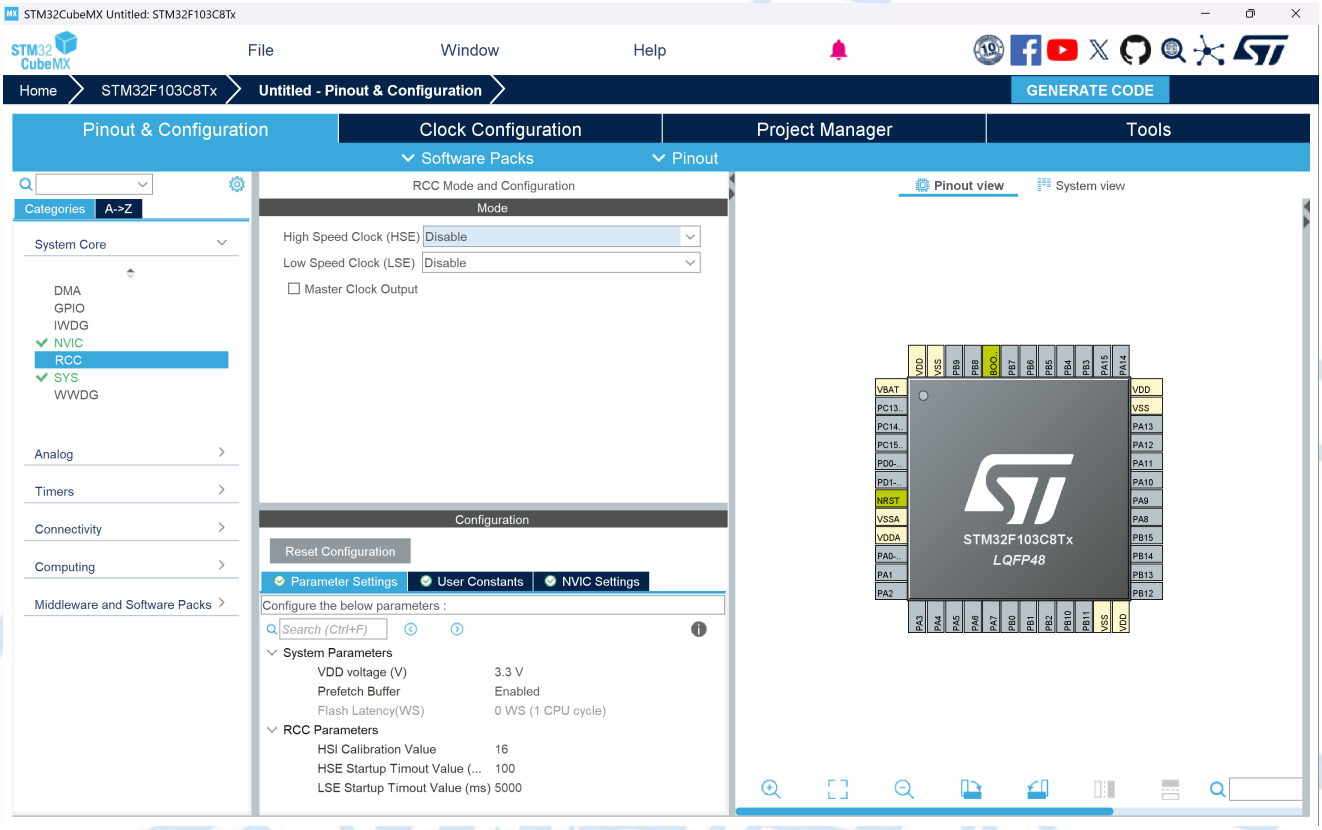
箭头指的这一栏中搜STM32f103C8t6




Cross Selector
Features
Block Diagram
Docs & Resources
CAD Resources
Datasheet
Buy
Start Project
STM32F1 Series
STM32F103C8T6
Mainstream Performance line, Arm Cortex-M3 MCU with 64 Kbytes of Flash memory, 72 MHz CPU, motor control, USB and CAN
ACTIVE
Product is in mass production
Unit Price for 10kU (US$) : 2.8462
LQFP 48 7x7x1.4 mm
The STM32F103xx medium-density performance line family incorporates the high-performance Arm® Cortex®-M3 32-bit RISC core operating at a 72 MHz frequency, high-speed embedded memories (Flash memory up to 128 Kbytes and SRAM up to 20 Kbytes), and an extensive range of enhanced I/Os and peripherals connected to two APB buses. All devices offer two 12-bit ADCs, three general purpose 16-bit timers plus one PWM timer, as well as standard and advanced communication interfaces: up to two I2Cs and SPIs, three USARTs, an USB and a CAN.
The devices operate from a 2.0 to 3.6 V power supply. They are available in both the –40 to +85°C temperature range and the –40 to +105 °C extended temperature range. A comprehensive set of power-saving mode allows the design of low-power applications.
MCUs/MPUs List: 2 items
Export
Commercial Pa... Part No Reference Marketing... Unit Price for 1... Board Package Flash RAM I/O Freq...
STM32F103C8T6 STM32F103C8 STM32F103C8... Active 2.8462 LQFP 48 7x7x1.4 mm 64 kBy... 20 kBytes 37 72 MHz
STM32F103C8T... STM32F103C8... Active 2.8462 LQFP 48 7x7x1.4 mm 64 kBy... 20 kBytes 37 72 MHz

2. 完成基础配置

接下来会进入到单片机图形化配置的界面

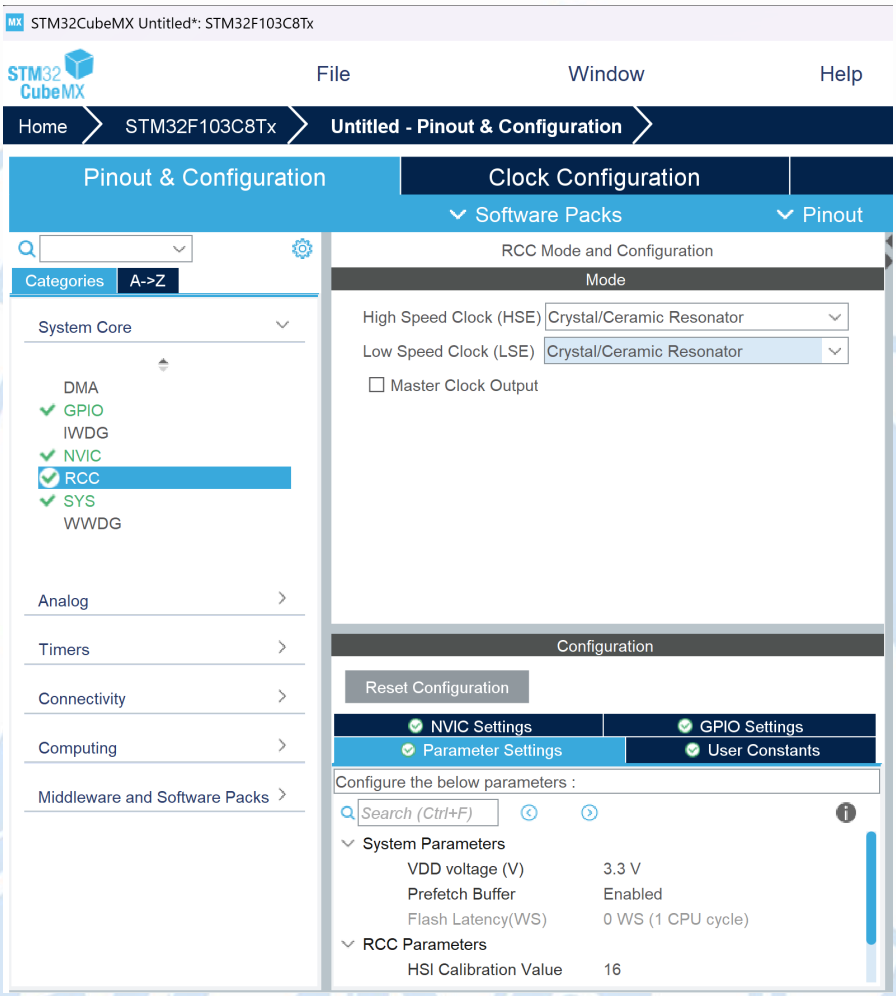


对于首次开发，主要有以下几项必做项

你们现在手上拿到的应该都是STM32f103c8t6的最小核心板，如果没有的话就去买江协的

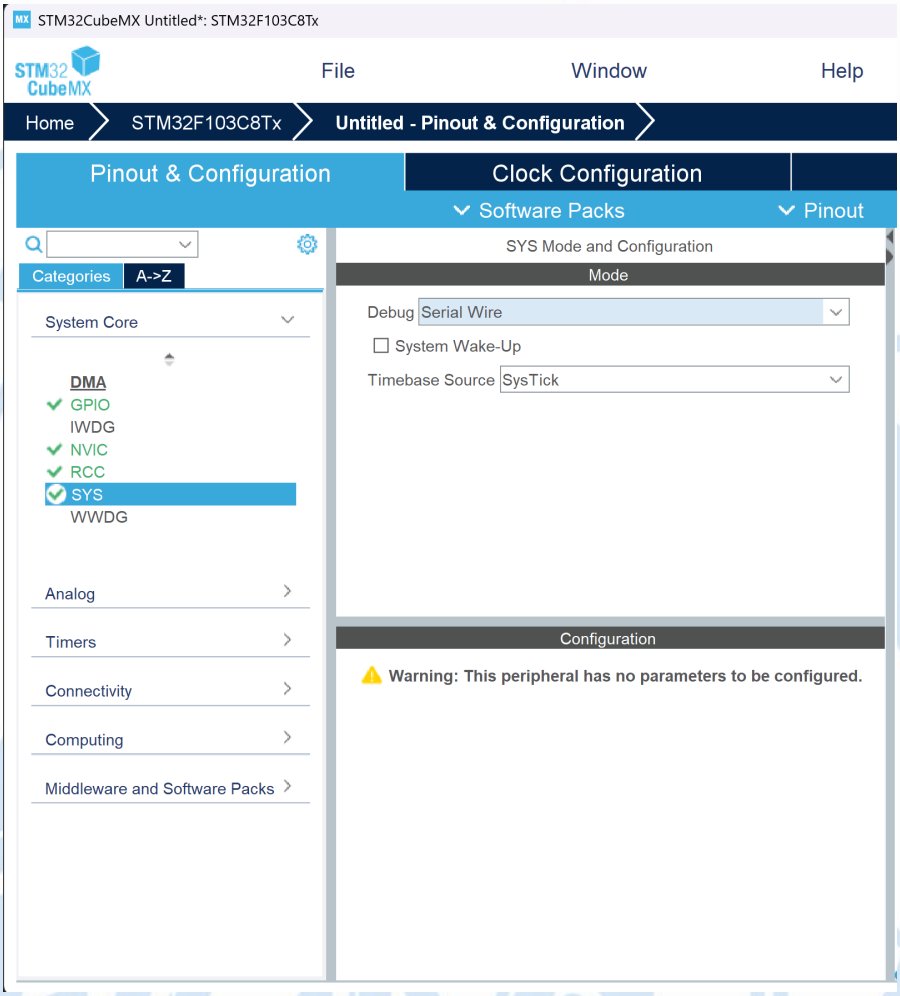
2.1 配置 RCC

在RCC一栏，右边的HSE和LSE都选择Crystal/Ceramic Resonator



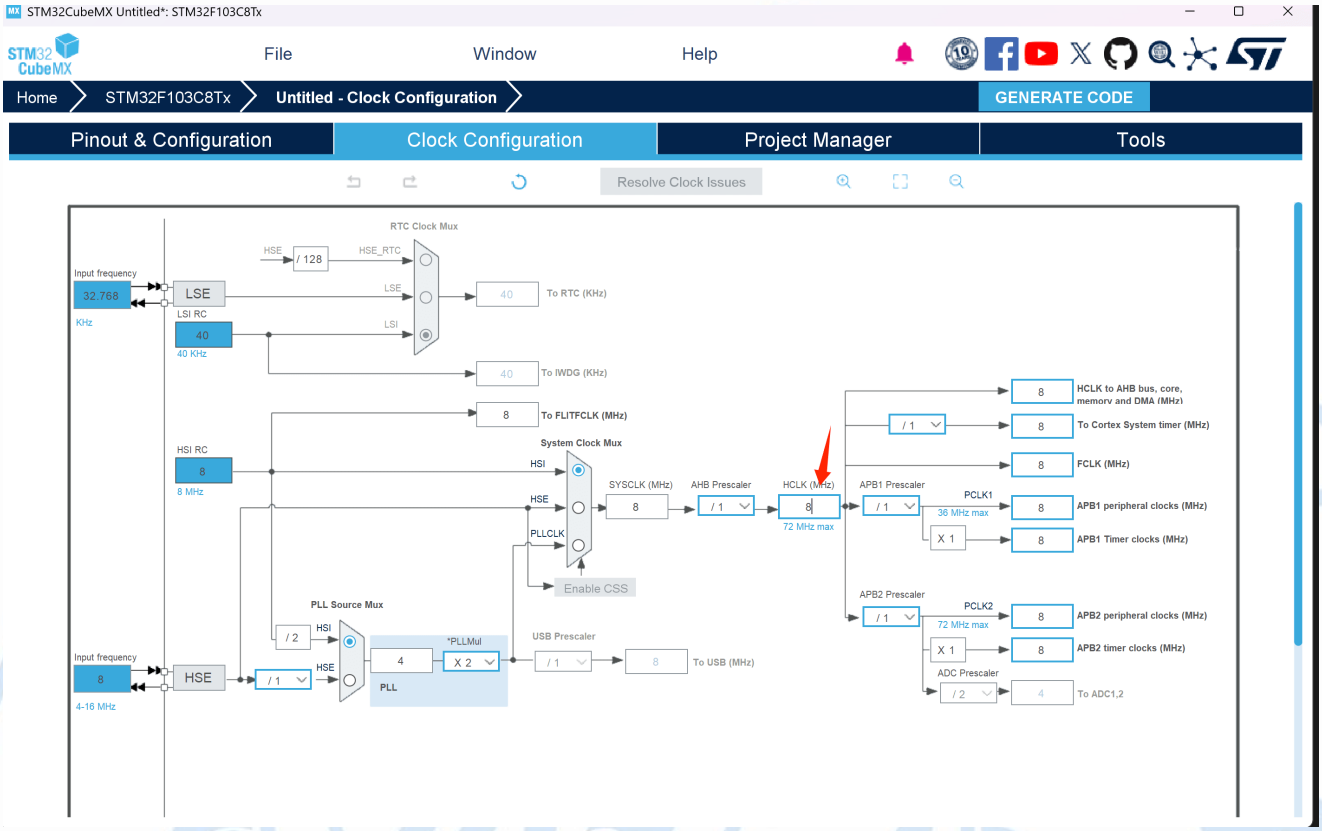
2.2 配置 SYS 调试接口

左边SYS这一栏，右侧选择Serial Wire



2.3 配置系统时钟

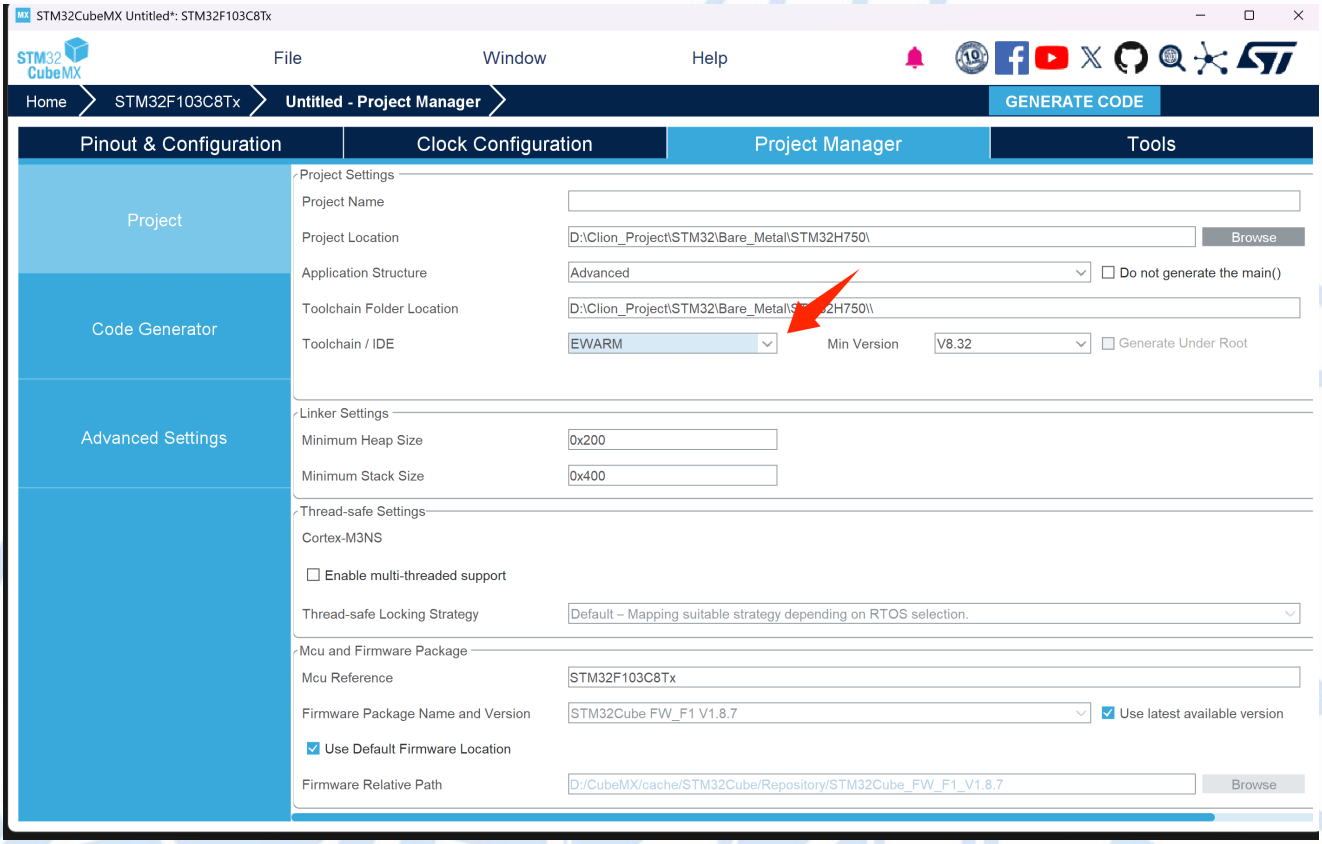
然后上面的Clock Configuration，把这里的8改成72

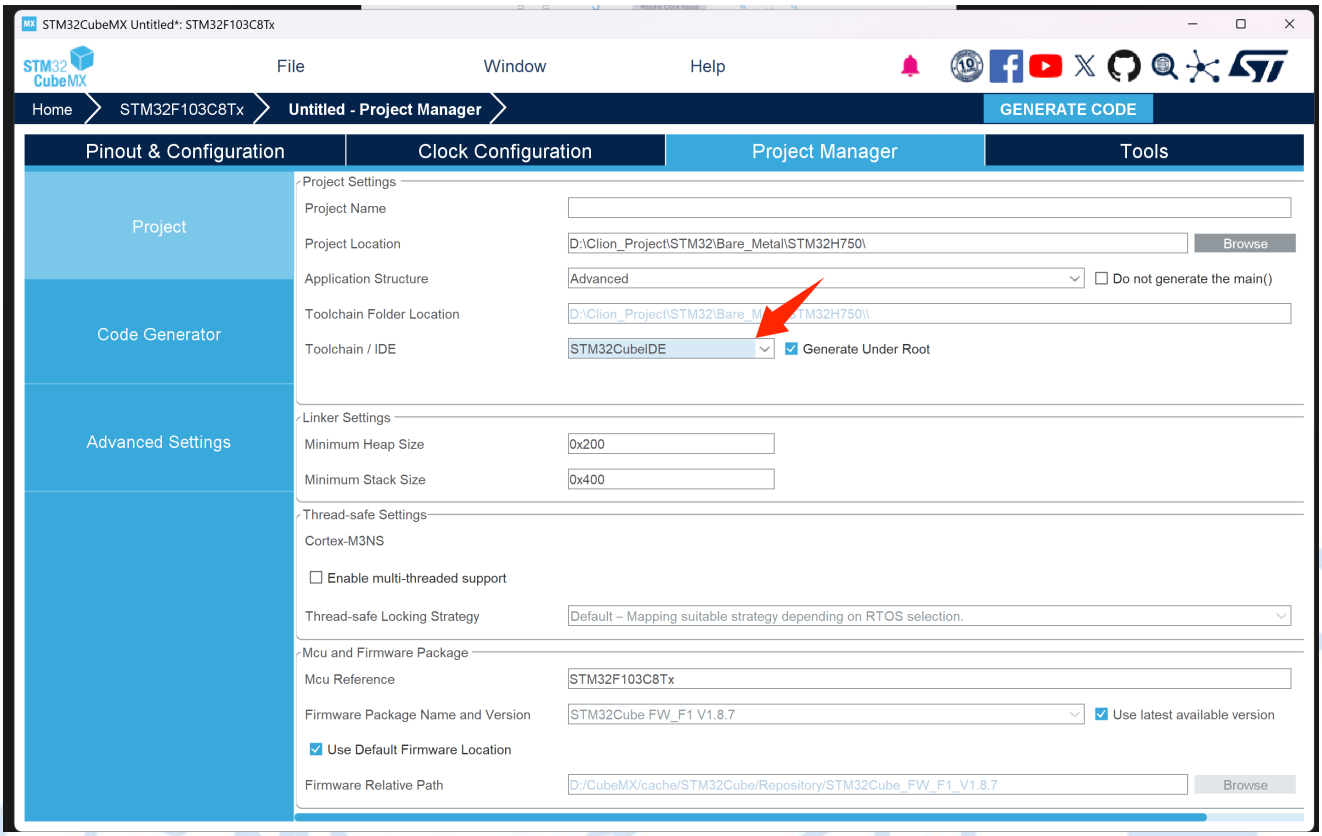


3. 配置并生成工程代码

3.1 选择工程工具链

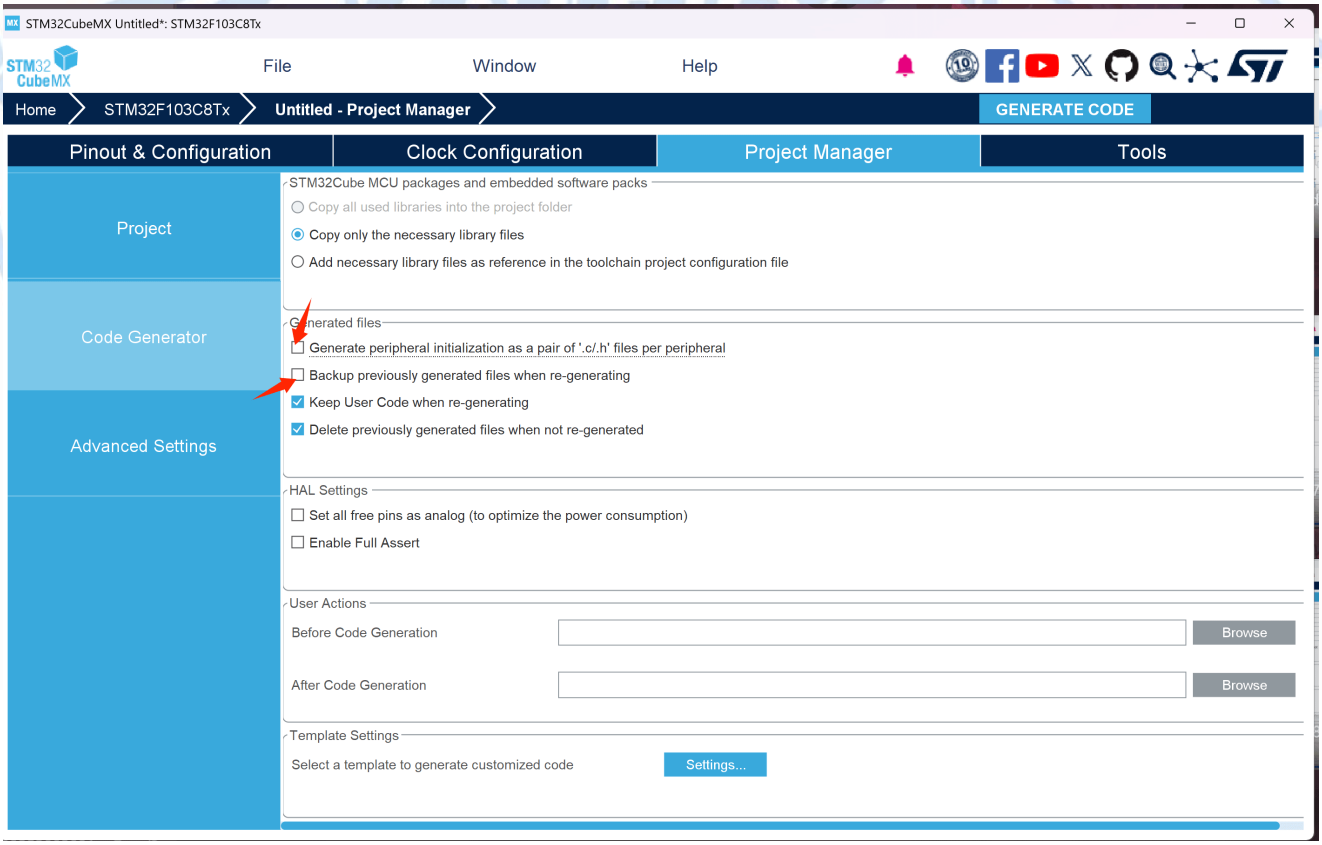
然后在Project Manager，这里的EWARM换成STM32CUBEIDE

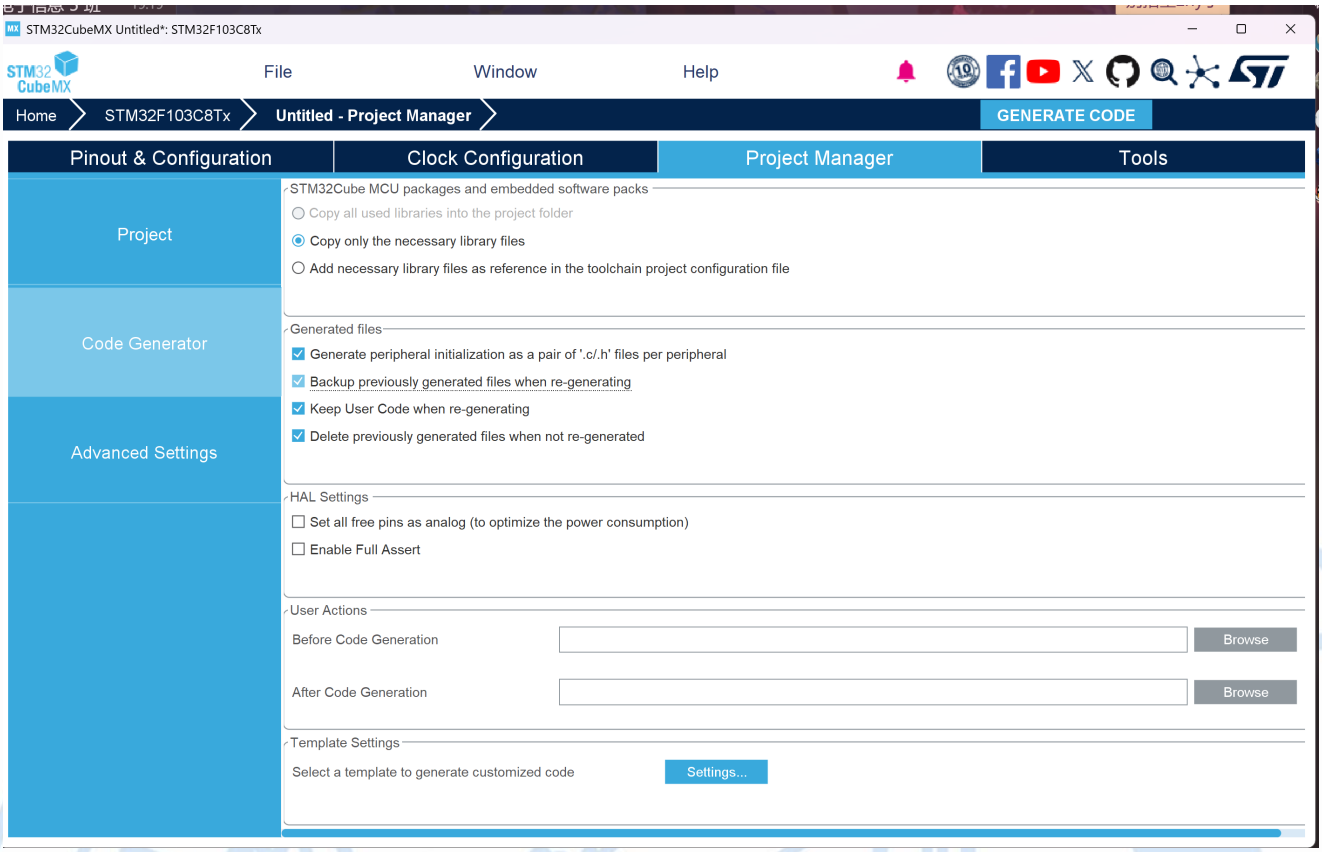


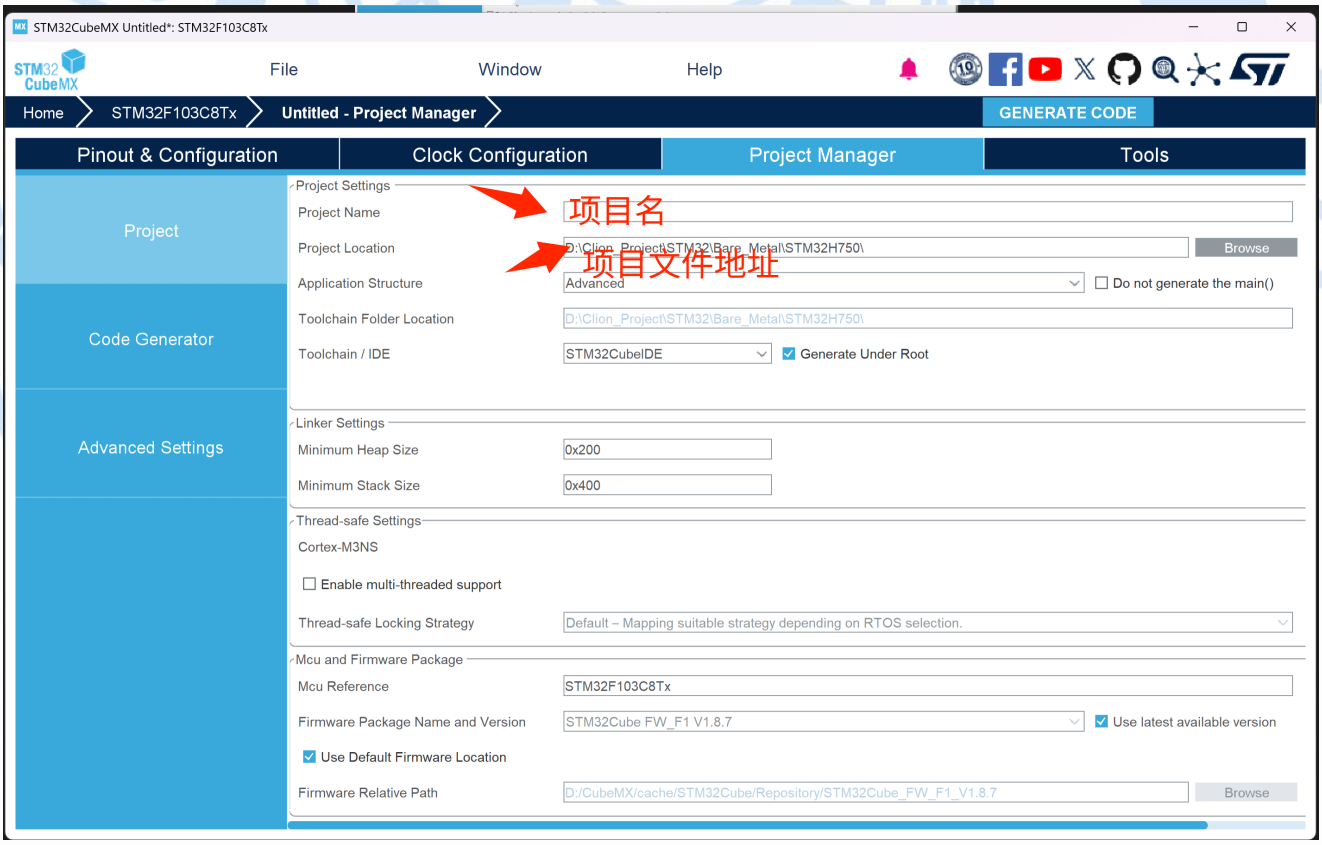


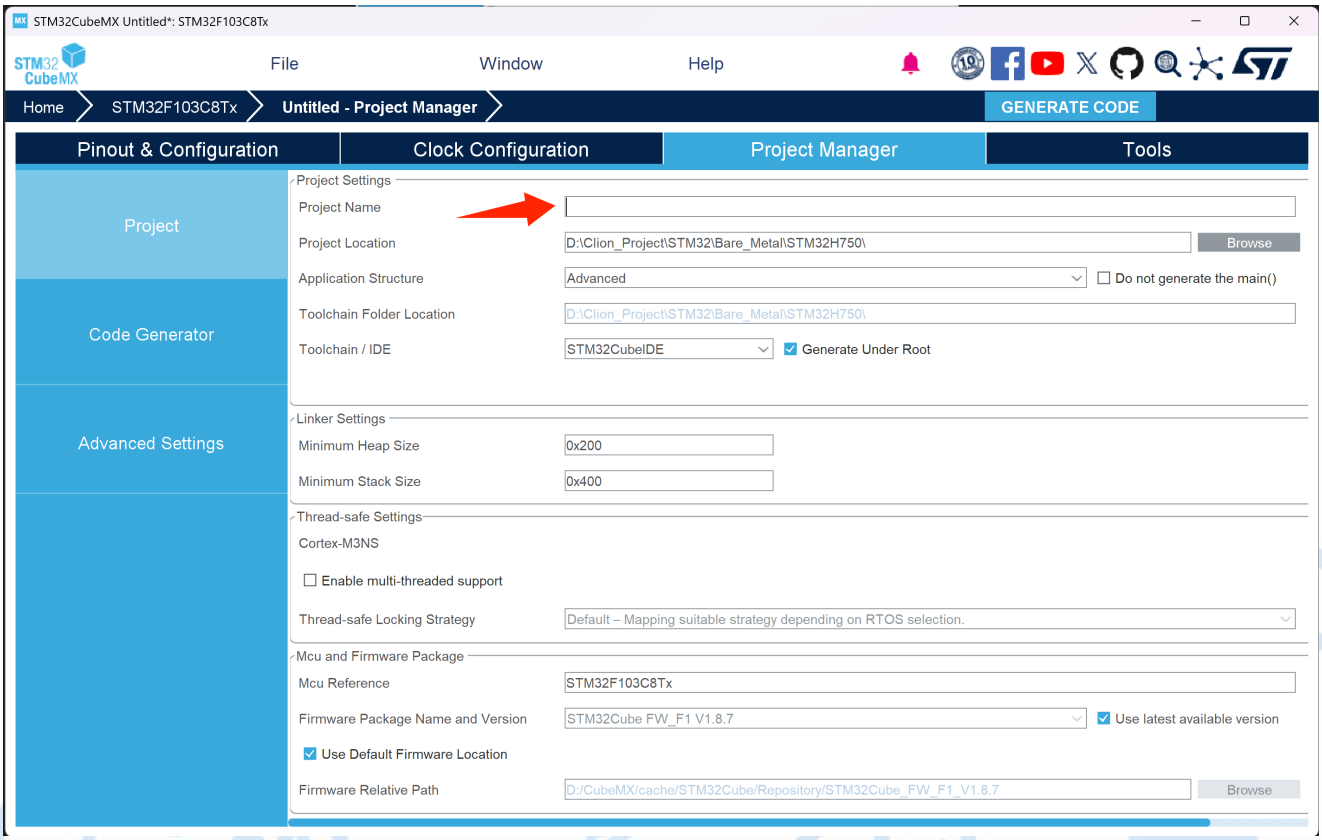
3.2 设置代码生成选项

然后在左侧的Code Generator中，把我箭头勾红的地方选上




STM32CubeMX Untitled*: STM32F103C8Tx
File
Window
Help
Home
STM32F103C8Tx
Untitled - Project Manager
GENERATE CODE
Pinout & Configuration
Clock Configuration
Project Manager
Tools
Project
Code Generator
Advanced Settings
STM32Cube MCU packages and embedded software packs
Copy all used libraries into the project folder
Copy only the necessary library files
Add necessary library files as reference in the toolchain project configuration file
Generated files
Generate peripheral initialization as a pair of '.c/.h' files per peripheral
Backup previously generated files when re-generating
Keep User Code when re-generating
Delete previously generated files when not re-generated
HAL Settings
Set all free pins as analog (to optimize the power consumption)
Enable Full Assert
User Actions
Before Code Generation
Browse
After Code Generation
Browse
Template Settings
Select a template to generate customized code
Settings...


STM32CubeMX Untitled*: STM32F103C8Tx
File
Window
Help
Home
STM32F103C8Tx
Untitled - Project Manager
GENERATE CODE
Pinout & Configuration
Clock Configuration
Project Manager
Tools
Project
Code Generator
Advanced Settings
Project Settings
Project Name
项目名
Project Location
D:\Clion_Project\STM32\Bare_Metal\STM32H750\
Browse
项目文件地址
Application Structure
Advanced
Do not generate the main()
Toolchain Folder Location
D:\Clion_Project\STM32\Bare_Metal\STM32H750\
Toolchain / IDE
STM32CubeIDE
Generate Under Root
Linker Settings
Minimum Heap Size
0x200
Minimum Stack Size
0x400
Thread-safe Settings
Cortex-M3NS
Enable multi-threaded support
Thread-safe Locking Strategy
Default – Mapping suitable strategy depending on RTOS selection.
Mcu and Firmware Package
Mcu Reference
STM32F103C8Tx
Firmware Package Name and Version
STM32Cube FW_F1 V1.8.7
Use latest available version
Use Default Firmware Location
Firmware Relative Path
D:/CubeMX/cache/STM32Cube/Repository/STM32Cube_FW_F1_V1.8.7
Browse



3.3 生成代码

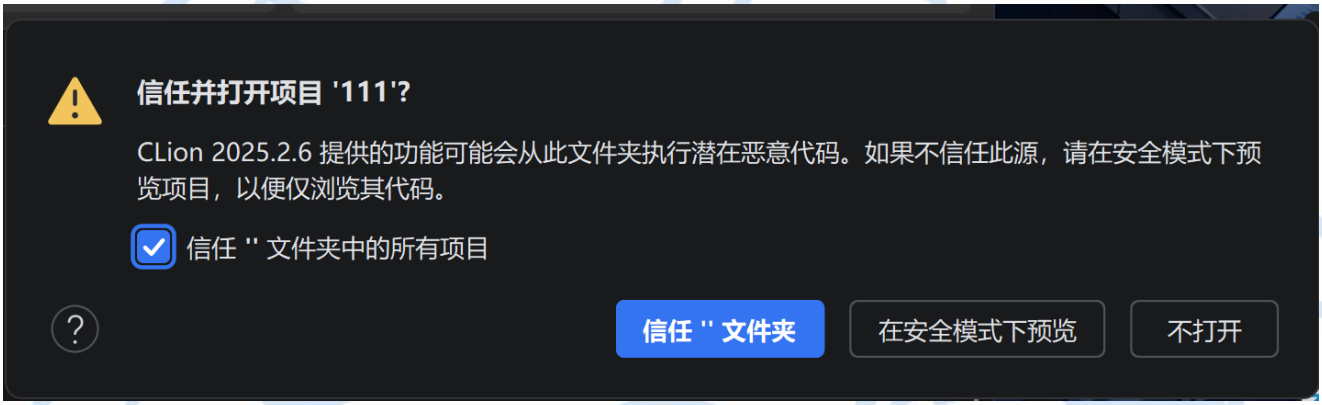
然后点右上角GENERATE CODE

4. 在 CLion 中打开并配置工程

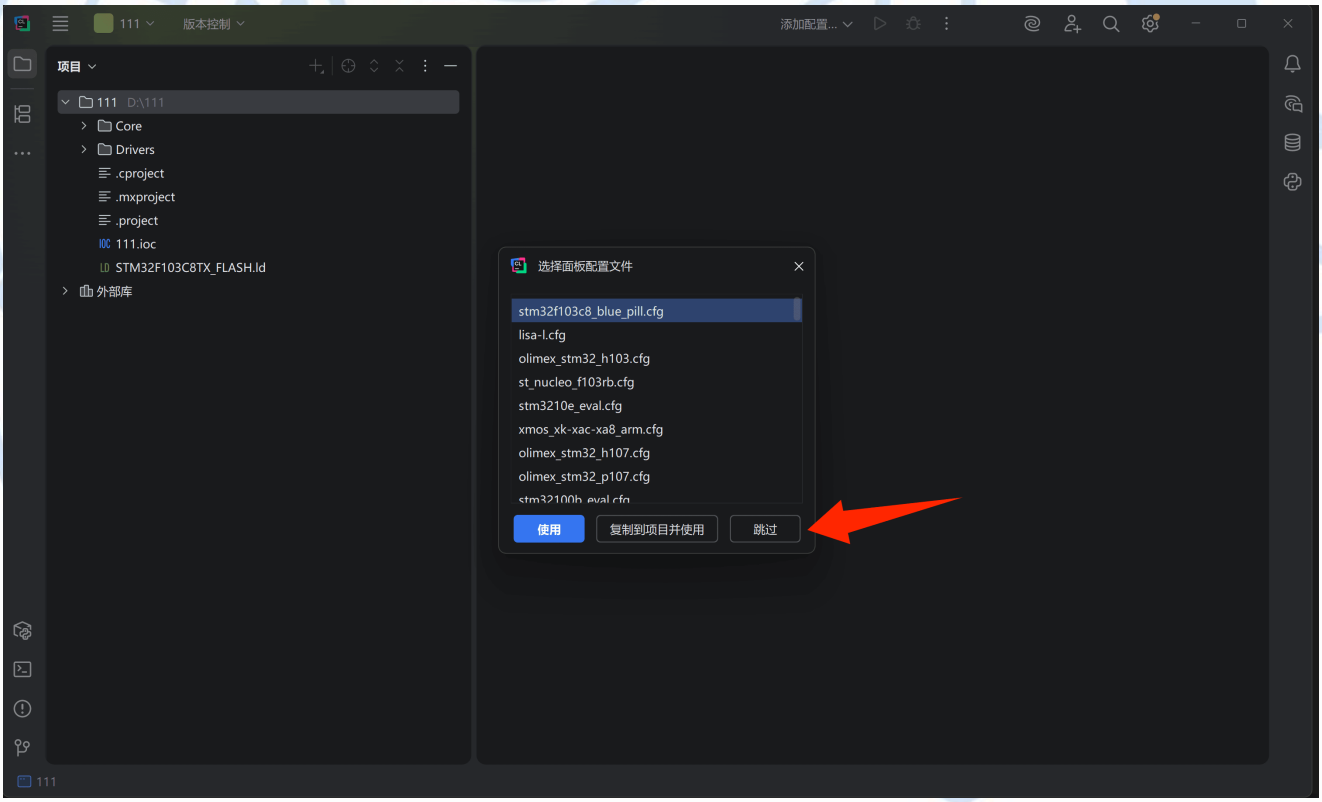
4.1 导入工程

生成项目文件后，将文件夹拖到Clion上

这里选信任

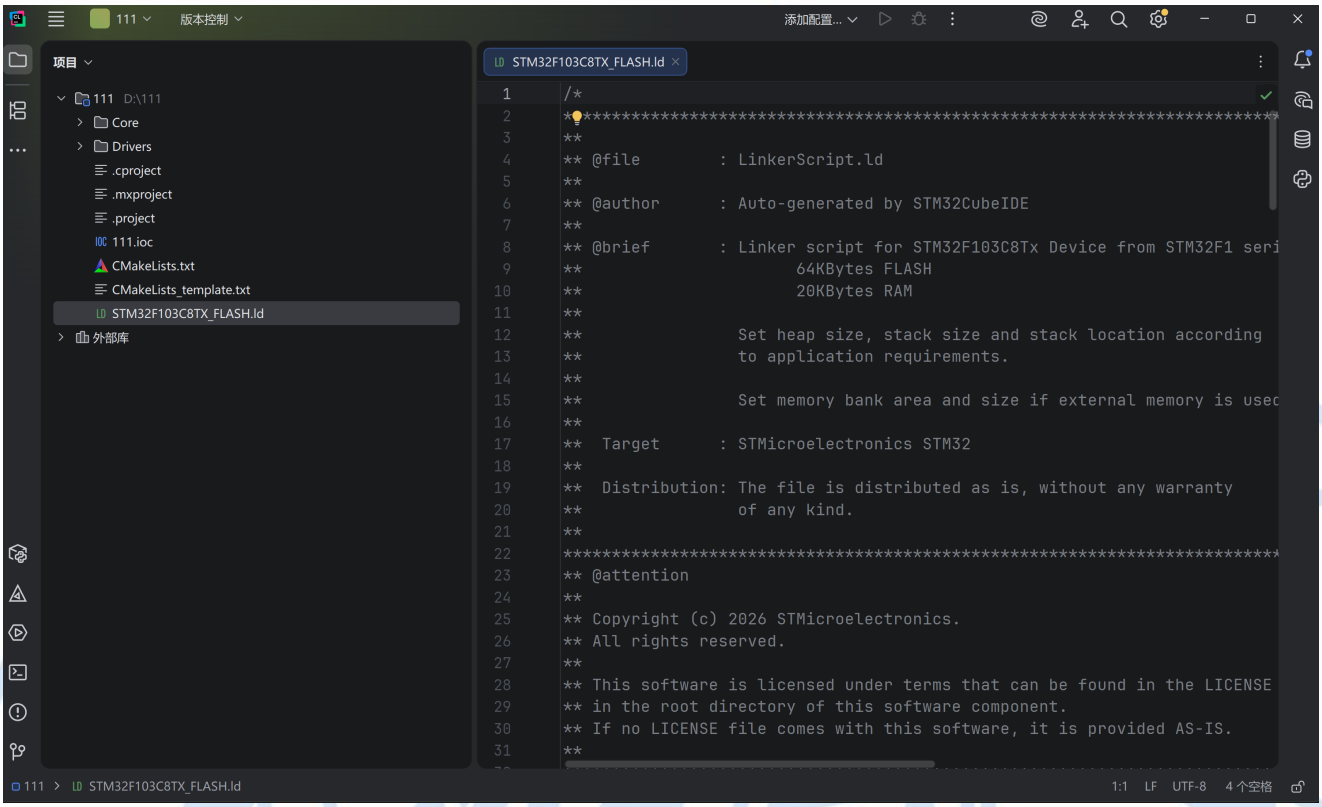


这里选跳过

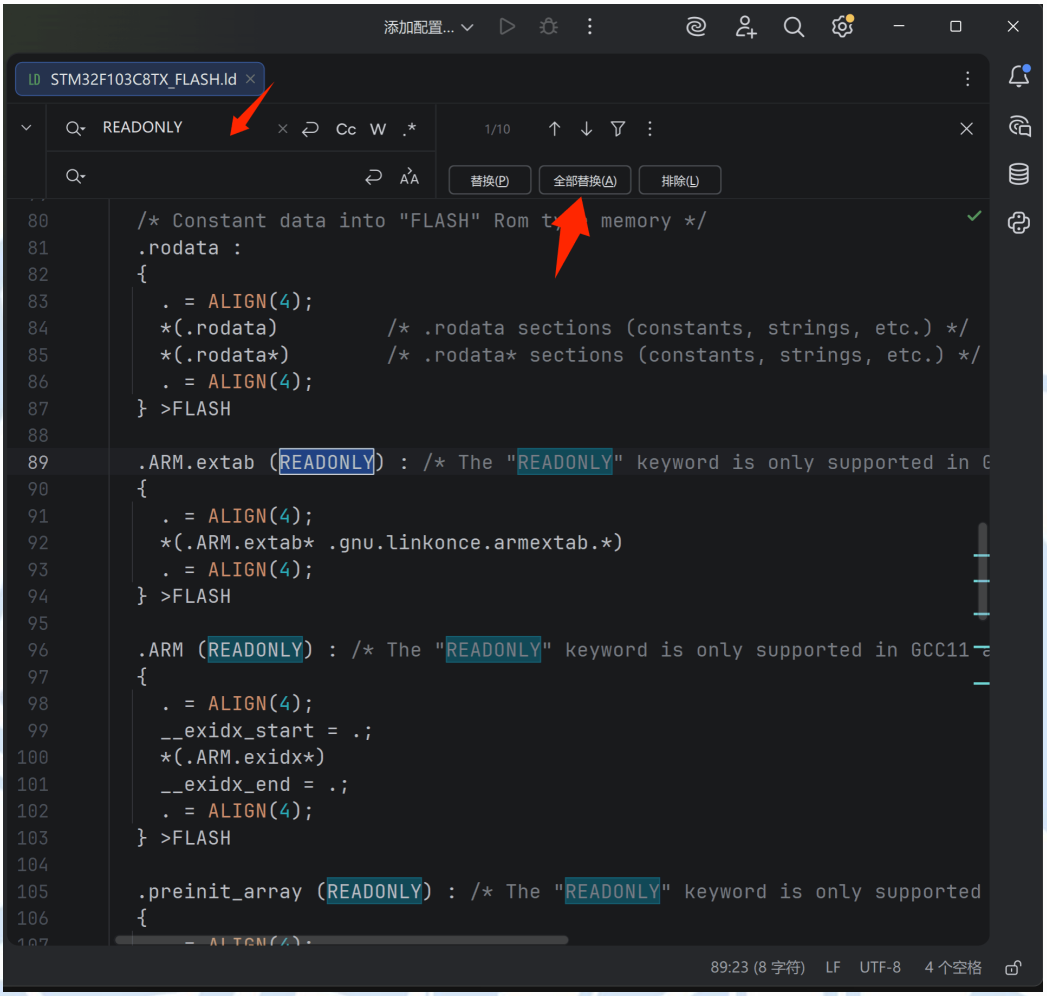


4.2 修改链接脚本

打开这里的.ld文件

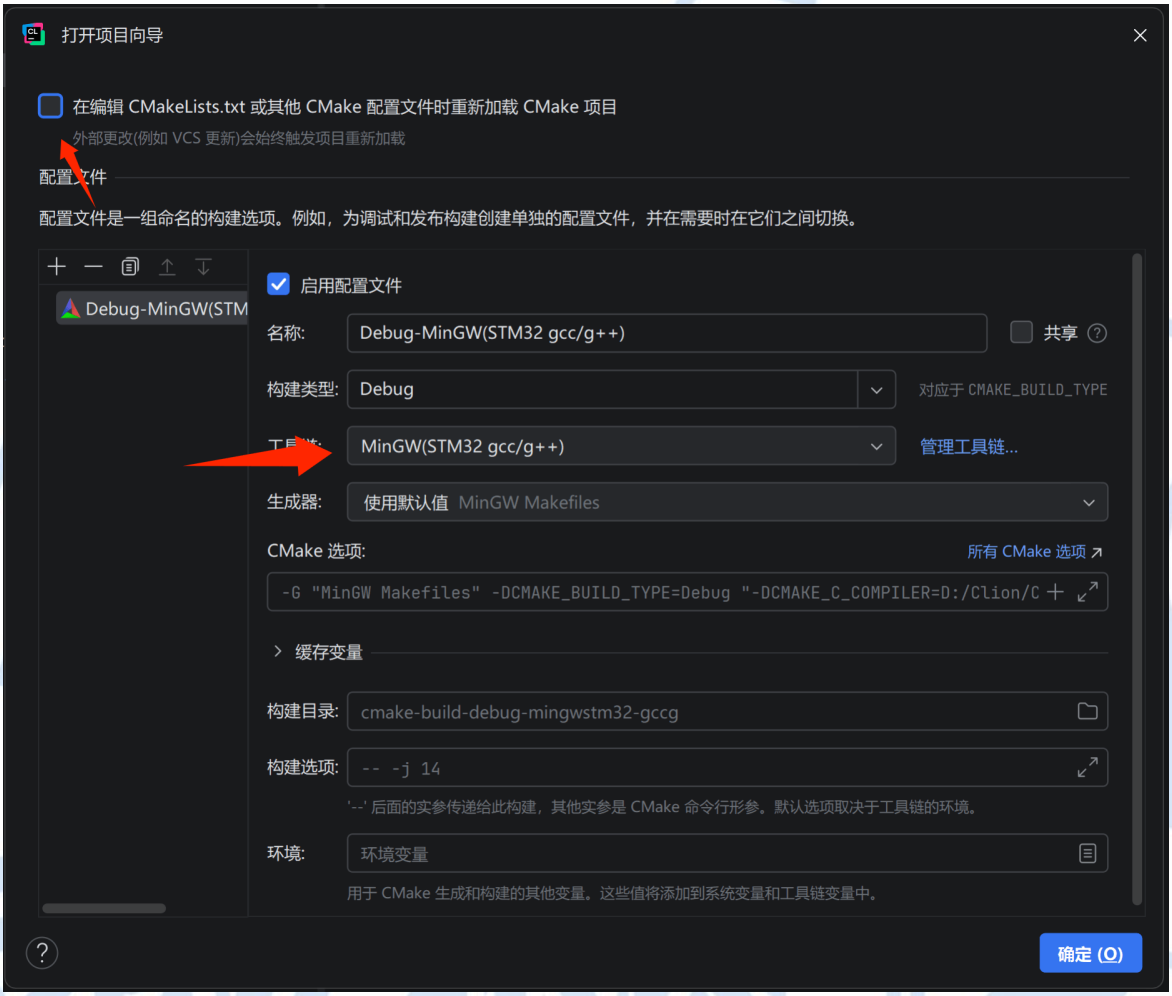


按Ctrl+R，在搜索栏输入READONLY，然后点下面的全部替换（相当于把ld文件中的READONLY这个字段全删了）

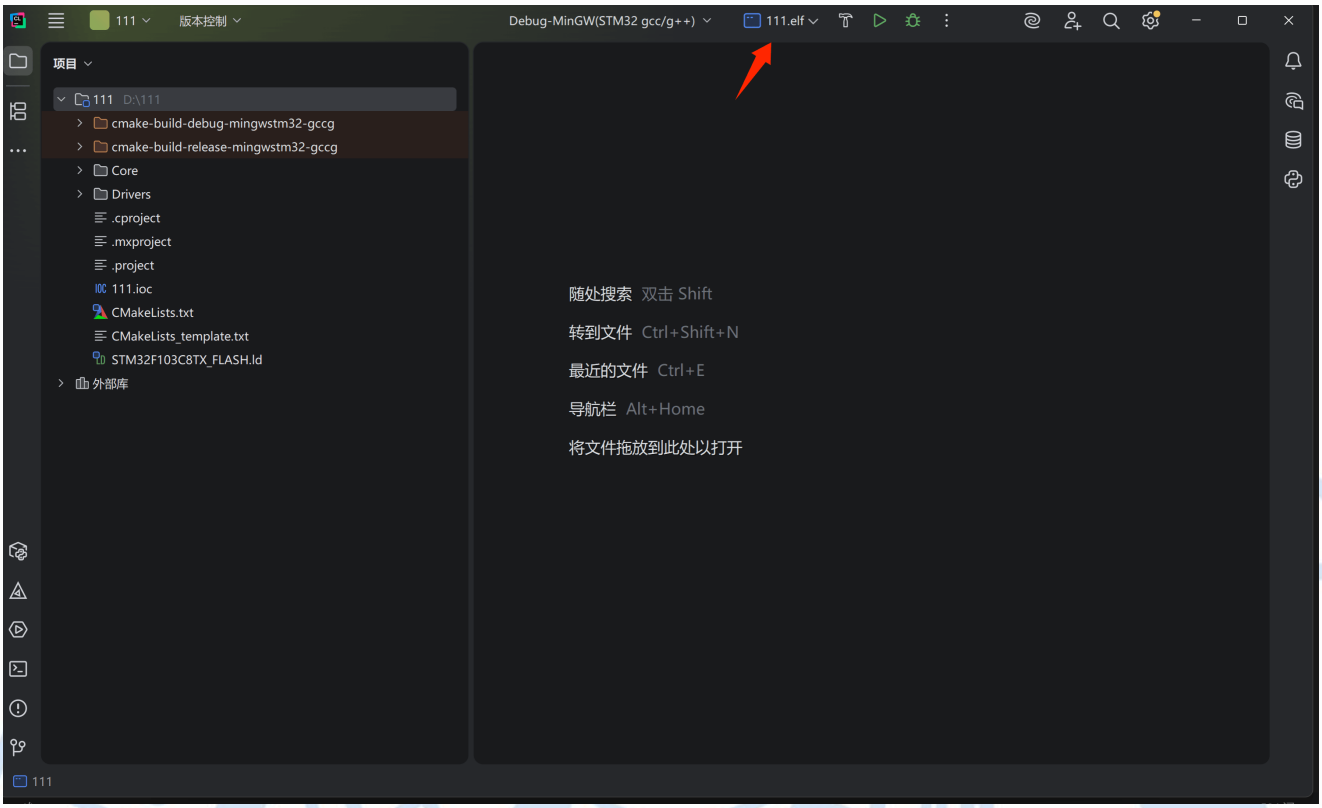


4.3 配置 MinGW 工具链

之后关闭Clion，然后再打开Clion，会进入这个界面，勾选左上角的选项，重点是配置工具链，从默认换成MinGW



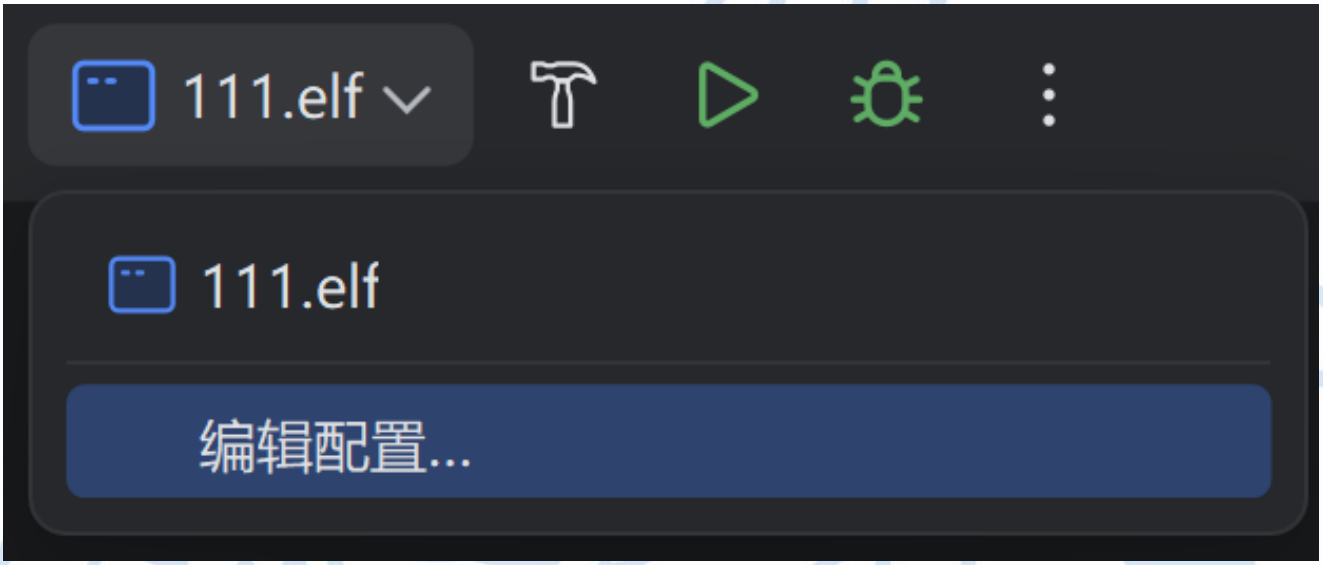
然后确定，返回IDE界面



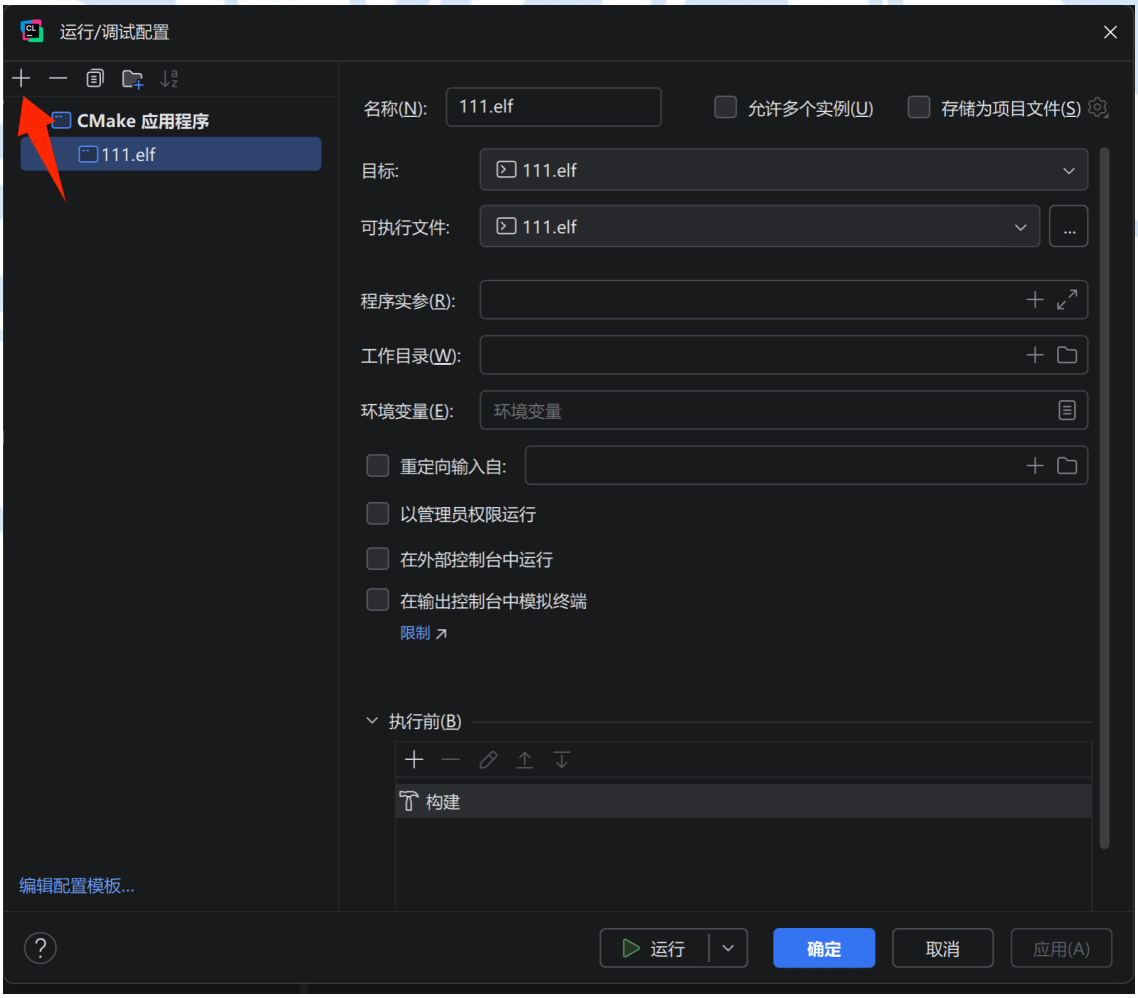
5. 配置 OpenOCD 并编译下载

5.1 添加运行配置

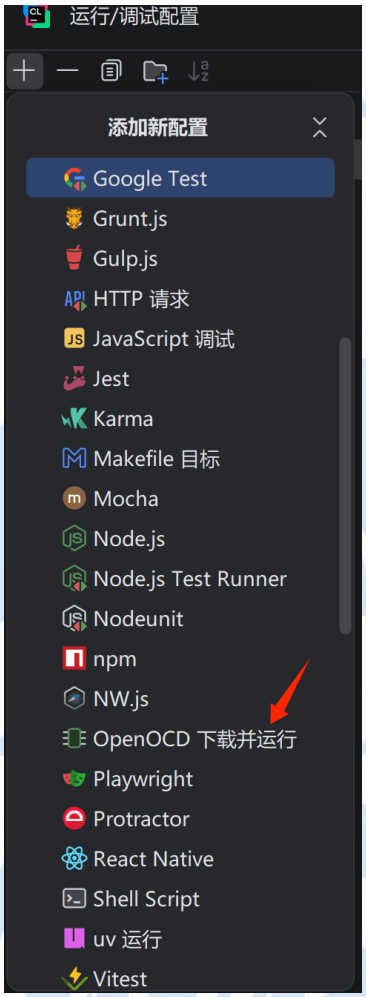
现在选择右上角的蓝色标，这个.elf，

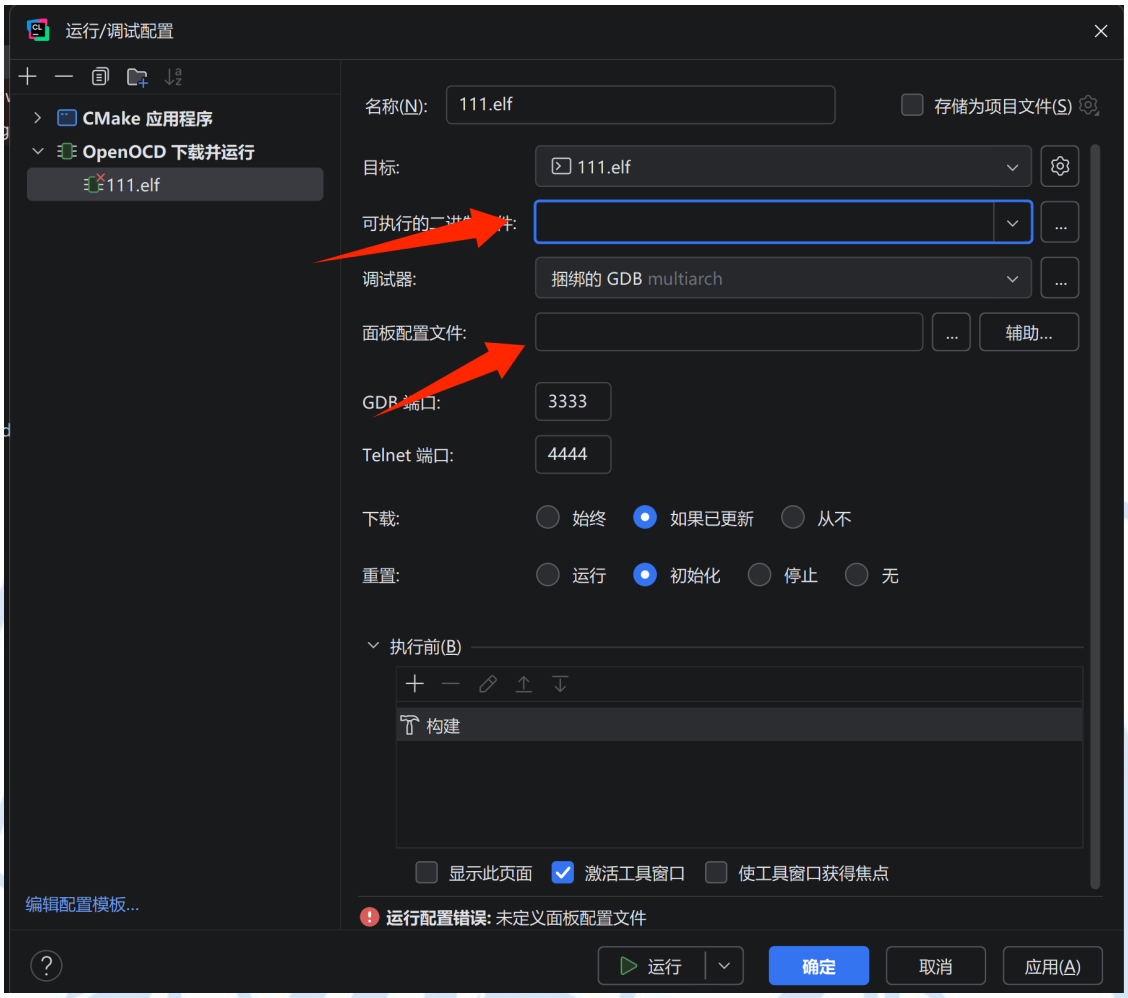


选这里编辑配置，然后左上角+号



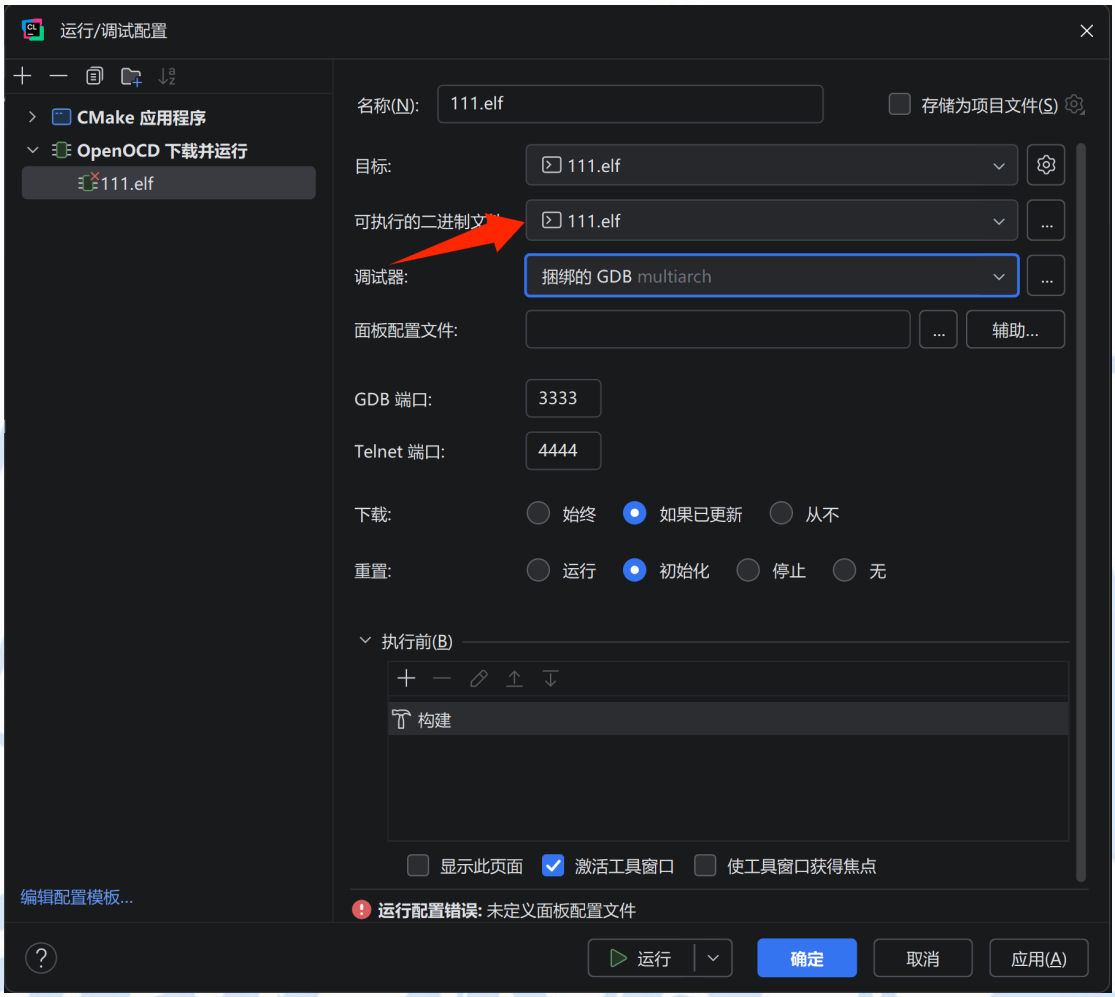
选OpenOCD下载并运行

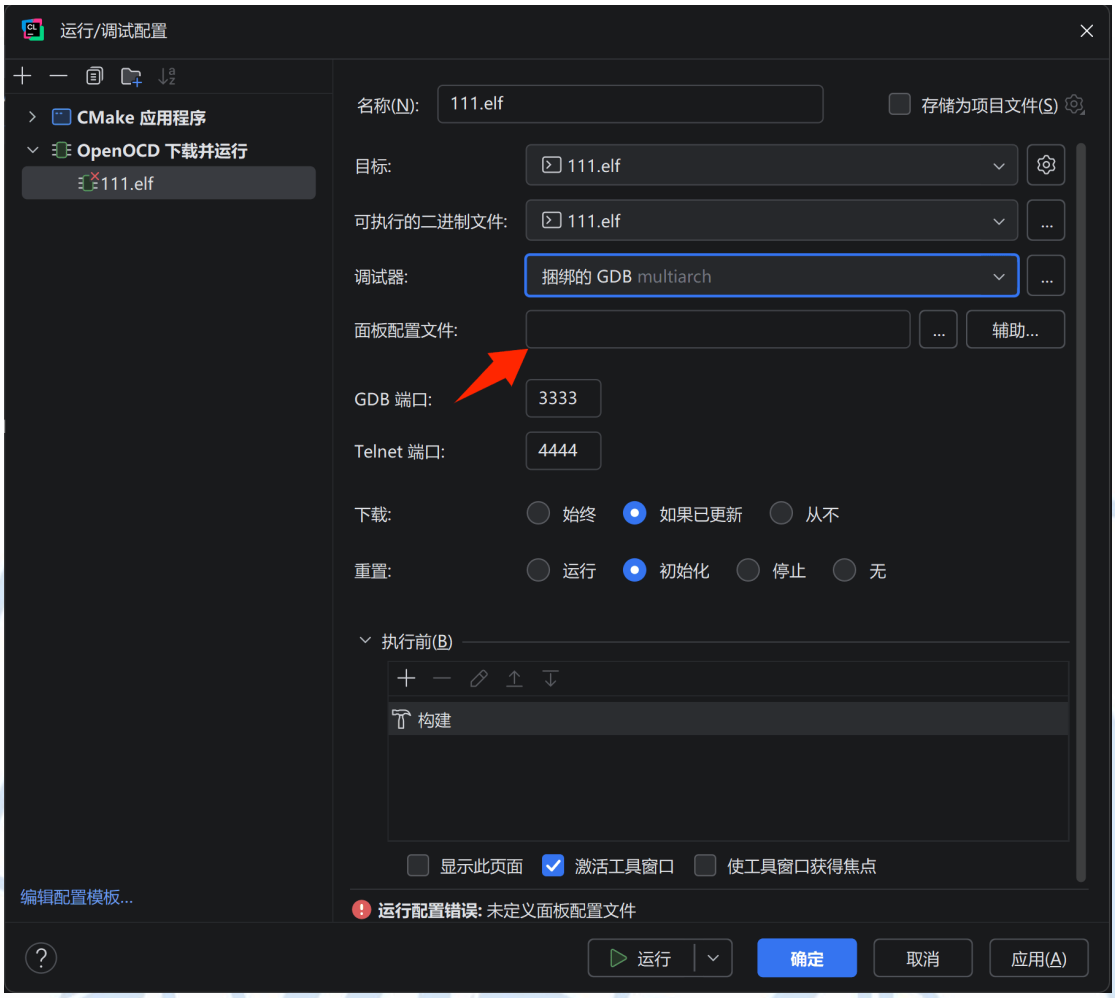



运行/调试配置
CMake 应用程序
OpenOCD 下载并运行
111.elf
名称(N): 111.elf
存储为项目文件(S)
目标: 111.elf
可执行的二进制文件:
调试器: 捆绑的 GDB multiarch
面板配置文件:
辅助...
GDB 端口: 3333
Telnet 端口: 4444
下载: 始终 如果已更新 从不
重置: 运行 初始化 停止 无
执行前(B)
构建
显示此页面 激活工具窗口 使工具窗口获得焦点
编辑配置模板...
运行配置错误: 未定义面板配置文件
运行
确定
取消
应用(A)

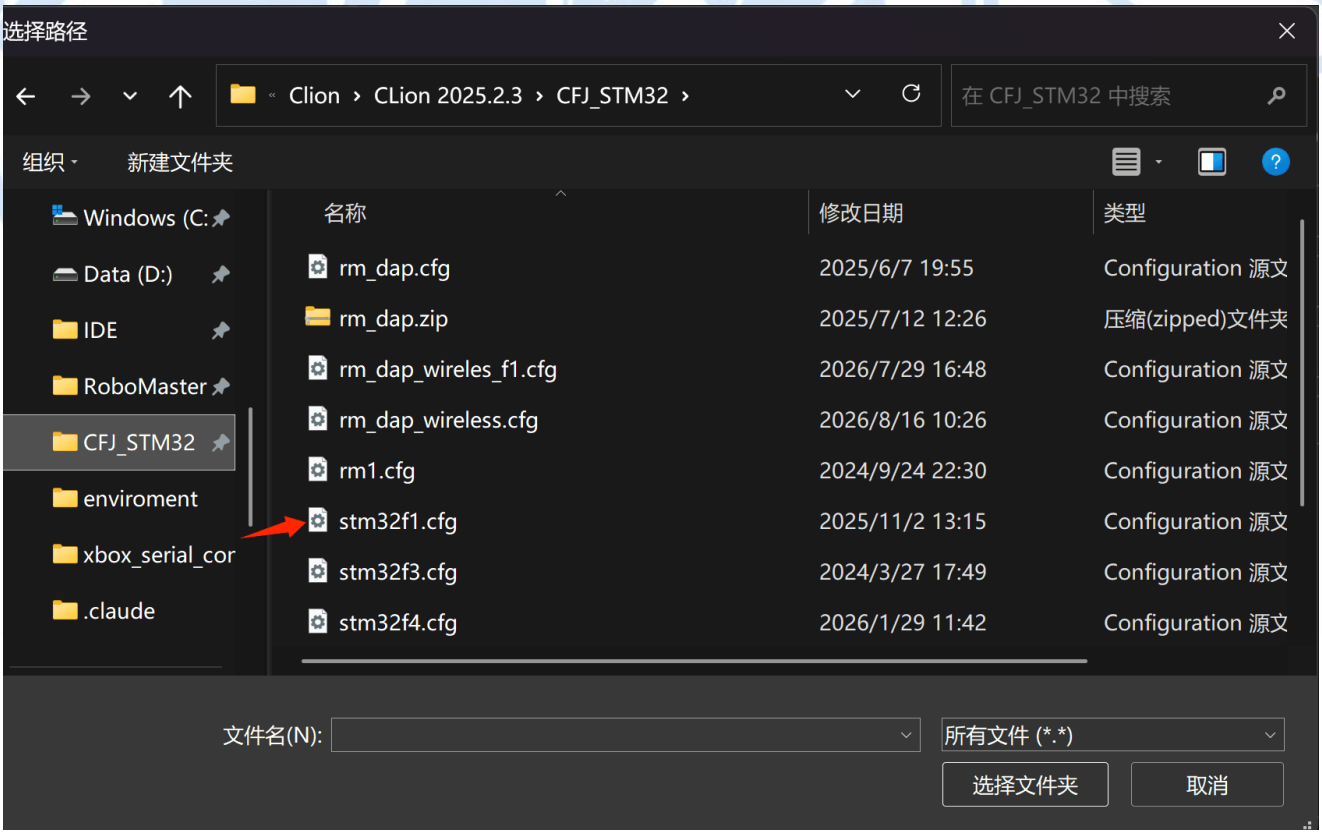
5.2 选择可执行文件与配置文件

可执行的二进制文件这里只有一个能选（.elf文件）





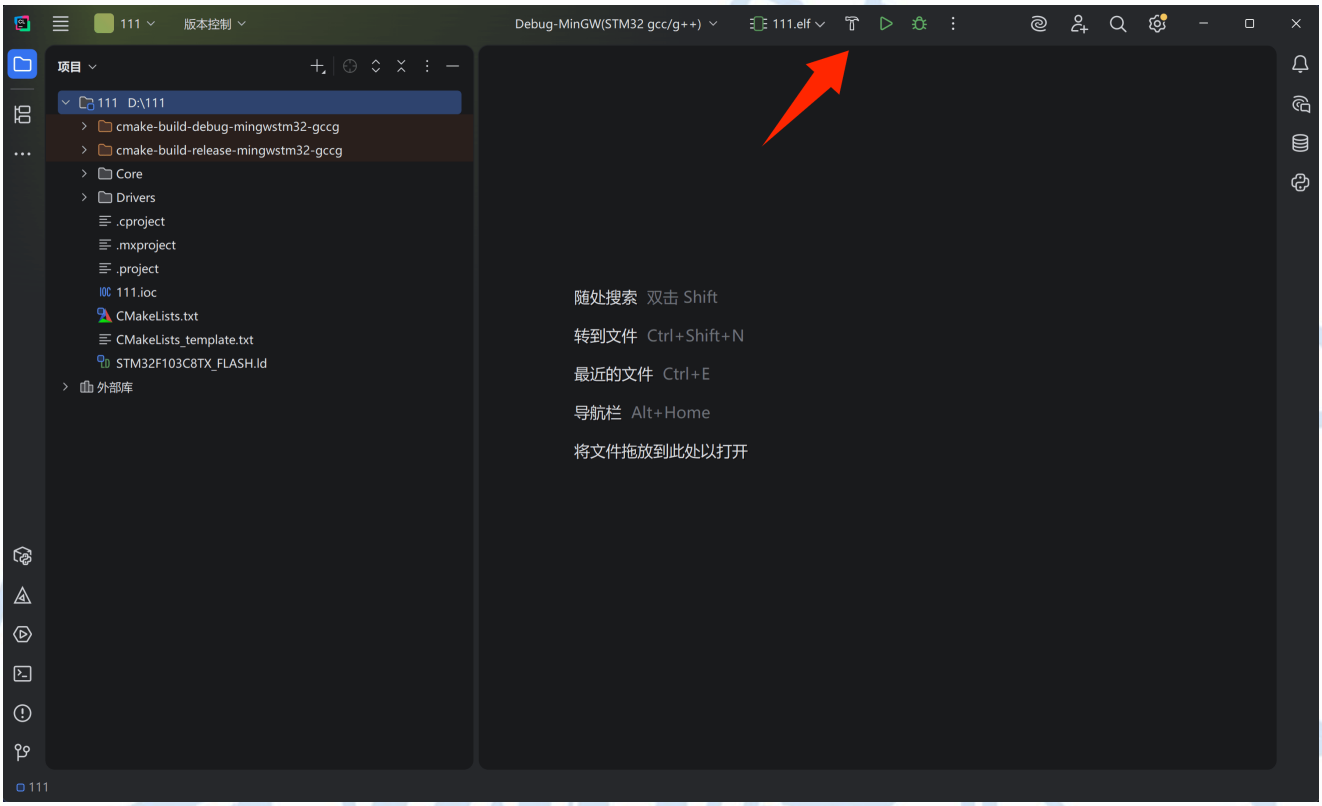
这里选之前发的CFJ_STM32中的stm32f1.cfg



选好了之后选下面的确定

5.3 编译与下载

这样就完成了一个项目的创建



这个小扳手是“编译” 绿色三角是编译并下载（烧录）